Trace: Claim-Level Evidence Verification in Retrieval-Augmented Phishing Assessments

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Abstract—Retrieval-augmented generation (RAG) can support phishing-message assessment, but retrieved evidence does not ensure that generated explanations are adequately supported. This study evaluates claim-level evidence verification in Trace, a RAG-based system for analyzing suspicious email and SMS messages. A paired experiment compared baseline and verified assessments on 200 messages, with 197 successful paired evaluations. Verification increased accuracy from 69.04% to 70.56% and F1-score from 94.18% to 95.79%, but the accuracy difference was not statistically significant (exact McNemar p = 0.453). The verifier classified 64 of 654 claims as having insufficient evidence. An exploratory, non-independent review suggested that some flagged claims remained in final assessments, underscoring the distinction between detecting evidence gaps and enforcing verification judgments. Verification also increased processing time. The results support further investigation of evidence-enforcement mechanisms and independent claim-level validation rather than a definitive claim of improved factual faithfulness.

Index Terms— retrieval-augmented generation; phishing detection; claim-level verification; factual faithfulness; trustworthy AI

I. INTRODUCTION

Phishing attacks commonly use deceptive emails and text messages to persuade recipients to disclose sensitive information, visit fraudulent websites, or take other harmful actions. Identifying these messages is an important cybersecurity task, particularly as attackers develop increasingly convincing forms of communication.

Large language models (LLMs), combined with retrieval-augmented generation (RAG), offer a way to analyze suspicious messages using information retrieved from external knowledge sources [1]. However, retrieving relevant information does not guarantee that every claim in a generated assessment is supported by the available evidence. A system may correctly classify a message as phishing while making unsupported statements about its sender, website, or intentions [2].

This creates a distinction between classification correctness and factual faithfulness. Classification correctness concerns whether the system identifies a message as phishing or legitimate. Factual faithfulness concerns whether the explanations supporting that decision are justified by the available evidence.

This study investigates claim-level evidence verification as a mechanism for improving the reliability of retrieval-augmented phishing assessments. Using Trace, a RAG-based message analysis system, the study compares assessments generated without verification against assessments subjected to an additional claim-verification stage.

Research Question

Does claim-level evidence verification improve the reliability of retrieval-augmented phishing assessments compared with the same system without verification?

The study examines two dimensions of reliability: phishing classification performance and the treatment of claims identified as insufficiently supported by retrieved evidence.

II. RELATED WORK

Retrieval-augmented generation combines retrieved information with language-model generation to support knowledge-intensive tasks [1]. However, research on hallucination in natural language generation identifies factual inconsistency and unsupported content as persistent reliability concerns [2]. Trace examines a narrower question: whether verifying individual claims against a fixed evidence set changes phishing assessments and the treatment of insufficiently supported explanations. The present evaluation does not establish superiority over other published phishing-detection systems.

III. METHODOLOGY

Research Design

This study used a paired experimental design to investigate the effect of claim-level evidence verification on retrieval-augmented phishing assessments.

Two conditions were evaluated:

Baseline condition: The system generated an initial assessment of a message using retrieved evidence, without additional claim verification.

Verified condition: The system examined claims from the initial assessment against the same message and retrieved evidence, then generated a revised assessment informed by the verification results.

Both conditions shared the same input message, retrieved evidence, and initial assessment. This design isolated the addition of the verification stage, although the verified condition also involved generating a revised response.

Dataset

The evaluation dataset consisted of 200 messages, evenly divided between 100 phishing and 100 legitimate examples. The messages were obtained from three source datasets: the Nazario phishing email corpus, the SMS Phishing dataset, and the Phishing Validation dataset.

The phishing examples comprised 80 emails from the Nazario corpus and 20 SMS messages labeled as smishing in the SMS

Phishing dataset. The legitimate examples comprised 50 emails labeled as safe in the Phishing Validation dataset and 50 SMS messages labeled as ham in the SMS Phishing dataset. Overall, the dataset contained 130 email messages and 70 SMS messages.

Before sampling, message text was cleaned by normalizing whitespace and removing leading and trailing spaces. Messages containing fewer than 30 characters were excluded. Duplicate messages were identified using case-insensitive comparisons of normalized text. Messages associated with conflicting labels were removed, and only the first occurrence of each remaining duplicate was retained.

Additional source-specific filtering excluded promotional spam messages from the SMS dataset and selected non-message records from the Nazario corpus.

Random sampling was performed separately within four predefined source-label groups using a fixed random seed of 42. The sampled messages were subsequently combined and randomly shuffled using the same seed. Each message was assigned a unique identifier, from trace_0001 to trace_0200.

The resulting dataset contained 100 phishing and 100 legitimate messages, with no duplicate message texts under the preprocessing procedure.

Of the 200 messages submitted for evaluation, 197 successfully completed both experimental conditions. Three messages encountered processing or validation errors and were excluded from the paired performance analysis. The final evaluation therefore included 98 phishing messages and 99 legitimate messages.

Experimental Configuration

The experiment used the `openai/gpt-4.1-mini` language model through OpenRouter.

For each input message, the retrieval system selected five evidence passages. These passages were shared between the baseline and verified conditions.

The verifier evaluated individual claims using three possible judgments:

• Supported: The claim was supported by the message or retrieved evidence.

• Contradicted: The available evidence conflicted with the claim.

• Insufficient: The available evidence did not adequately establish the claim.

The verified assessment was generated using these judgments.

Classification and Evaluation

The system's risk assessments were mapped to predictions as follows:

• High risk: phishing

• Low risk: legitimate

• Medium risk: abstention

• Inconclusive risk: abstention

Classification performance was evaluated using accuracy, precision, recall, and F1-score. Abstentions were tracked separately and counted as incorrect for overall accuracy.

The study also examined changes in predicted risk levels, processing times, and the verifier's claim-level judgments.

To investigate whether verification warnings influenced final explanations, claims marked as insufficient were reviewed and categorized according to whether they were removed, qualified, or retained in the verified assessment.

Failed Evaluations

Three of the 200 attempted evaluations failed during assessment processing, resulting in a completion rate of 98.5%.

Two failures (trace_0024 and trace_0176) involved missing explanation fields. The third (trace_0031) occurred when the verifier referenced an unknown evidence identifier.

These cases were excluded from the paired classification and claim-level analyses. Their exclusion represents a limitation of the evaluation because the reported results describe successfully completed assessments rather than all attempted inputs.

IV. RESULTS AND ANALYSIS

Classification Performance

The paired evaluation included 197 successfully completed assessments. The baseline system correctly classified 136 messages (69.04%), while the verified system correctly classified 139 messages (70.56%), representing an improvement of 1.52 percentage points.

Precision increased from 97.80% to 98.91%, recall from 90.82% to 92.86%, and F1-score from 94.18% to 95.79%.

The baseline system abstained on 59 messages, compared with 57 for the verified system. Classification coverage consequently increased from 70.05% to 71.07%. [TABLE_RESULTS]

TABLE I. CLASSIFICATION PERFORMANCE ON 197 PAIRED EVALUATIONS

Metric	Baseline	Verified
Accuracy	69.04%	70.56%
Precision	97.80%	98.91%
Recall	90.82%	92.86%
F1-score	94.18%	95.79%
Abstentions	59	57
Coverage	70.05%	71.07%

Changes in Classification Decisions

Verification changed the predicted risk category for 8 of the 197 messages (4.06%). The remaining 189 assessments (95.94%) retained their original risk categories.

Of the eight changes, five corrected previously incorrect assessments, two changed correct assessments into incorrect ones, and one changed an incorrect assessment into another incorrect outcome.

McNemar's exact test [3] was applied to the paired classification outcomes to assess whether the observed difference was statistically significant. The test produced a two-sided p-value of 0.453, indicating that the improvement was not statistically significant at the 0.05 level.

Claim-Level Verification

Across the 197 completed assessments, the verifier evaluated 654 individual claims. Of these, 590 (90.21%) were classified as supported, 64 (9.79%) as having insufficient evidence, and none as contradicted.

An exploratory, assistant-generated semantic review (not independently validated) of the 64 claims marked as insufficient examined whether each claim was removed, qualified, or retained in the final verified assessment.

The review classified 27 claims (42.19%) as removed, 14 (21.88%) as qualified, and 23 (35.94%) as retained without adequate qualification.

These findings suggest that the verification stage identified evidence gaps but did not consistently prevent insufficiently supported assertions from appearing in final responses. Because these classifications have not been independently validated, they should not be interpreted as definitive factual error rates.

Processing Time

The baseline system required an average of 3.17 seconds to generate an assessment. The additional claim-verification and revision stage required an average of 7.05 seconds, resulting in a total average processing time of 10.22 seconds per message.

This indicates that claim-level verification introduced substantial additional computational and response-time overhead.

Summary of Findings

The results indicate that claim-level evidence verification produced a modest improvement in observed phishing classification performance. However, the improvement was not statistically significant, and the majority of classification decisions remained unchanged.

The claim-level analysis also revealed limitations in the system's ability to enforce verification judgments. Although many claims identified as insufficiently supported were removed or qualified, some persisted in final assessments without adequate qualification.

Together, these findings suggest that claim-level verification may contribute to more reliable phishing assessments, but its effectiveness depends on how consistently verification results are incorporated into the final response.

V. DISCUSSION AND LIMITATIONS

Interpretation of Classification Results

The results indicate that claim-level evidence verification produced a modest improvement in phishing classification performance. Accuracy increased from 69.04% to 70.56%, while F1-score increased from 94.18% to 95.79%. However, verification changed only eight of the 197 classification decisions, and the observed improvement was not statistically significant.

One possible explanation is that the initial assessments already contained sufficient information to establish a risk category in most cases. Consequently, the verification stage may have had limited influence on the overall classification decision.

These findings suggest that claim-level verification should not be evaluated solely by its ability to improve classification accuracy. Its potential value also lies in improving the evidential justification of generated assessments.

Evidence Verification and Factual Faithfulness

The claim-level analysis revealed an important distinction between detecting insufficient evidence and preventing unsupported assertions.

The verifier identified 64 claims as having insufficient evidence. A provisional review found that 27 were removed, 14 were qualified, and 23 remained without adequate qualification in the final assessments.

This suggests that identifying an evidence gap does not necessarily ensure that the final assessment responds appropriately to that finding.

A possible explanation is that the final response generator retains information from the original assessment even after the verifier identifies weaknesses in its evidential support. This highlights a limitation of relying on language model instructions alone to enforce verification decisions.

The findings motivate further investigation into mechanisms that explicitly control whether claims can appear in final assessments based on their verification status.

Classification Accuracy Versus Explanation Reliability

An important implication of this study is that classification correctness and explanation reliability are related but distinct evaluation objectives.

A system may correctly classify a message as phishing while presenting claims that are not adequately supported by the available evidence. Conversely, a system may produce an appropriately cautious explanation without arriving at the correct classification.

Therefore, improvements in classification metrics do not necessarily establish improvements in factual faithfulness.

Evaluating evidence-aware AI systems requires examining both the correctness of their decisions and the support provided for their explanations.

Limitations

Several limitations should be considered when interpreting the findings.

First, the evaluation included 197 successfully completed assessments from a dataset of 200 messages. Although the dataset contained both phishing and legitimate examples across email and SMS formats, its size and composition limit the generalizability of the findings.

Second, the experiment used a single language model, `openai/gpt-4.1-mini`. The results may differ with other models, prompting strategies, or verification architectures.

Third, the study used five retrieved evidence passages per message. The quality and relevance of these passages may have influenced both the initial assessments and the verifier's judgments. The experiment did not systematically vary retrieval quality.

Fourth, the claim-level analysis relied on judgments produced by the system's own verifier. These judgments were not independently validated against expert annotations. Consequently, a claim marked as insufficient does not necessarily represent a factual error.

Fifth, the subsequent review of insufficiently supported claims was provisional. The categories of removed, qualified, and retained require independent validation before they can support stronger

conclusions about factual faithfulness.

Finally, the verified condition required additional processing time. The average verification and revision stage took 7.05 seconds, compared with 3.17 seconds for baseline generation. This overhead may affect the practicality of verification in applications requiring rapid responses.

Implications for Future Research

The findings suggest several directions for future investigation.

One direction is the development of stricter evidence-enforcement mechanisms that prevent insufficiently supported claims from being presented as established facts.

Another is evaluating the verifier against independently annotated claims to distinguish correct evidence judgments from overly cautious or incorrect verification decisions.

Future studies could also investigate how retrieval quality, evidence sufficiency, and model selection influence verification effectiveness.

Finally, larger evaluations across different datasets and communication formats would help determine whether the observed classification and explanation patterns generalize beyond the present experiment.

VI. CONCLUSION

This study investigated whether claim-level evidence verification improves the reliability of retrieval-augmented phishing assessments. Using Trace, an evidence-aware phishing assessment system, a paired evaluation compared baseline assessments with assessments subjected to an additional claim-verification stage.

Across 197 successfully completed evaluations, verification increased classification accuracy from 69.04% to 70.56% and F1-score from 94.18% to 95.79%. However, only eight classification decisions changed, and the observed improvement was not statistically significant (McNemar's exact test, $p = 0.453$). These results provide limited evidence that the verification stage improves phishing classification performance under the evaluated conditions.

The claim-level analysis revealed a separate challenge. Of 654 evaluated claims, the verifier identified 64 as having insufficient evidence. A provisional review found that 41 of these claims were removed or qualified in the final assessments, while 23 remained without adequate qualification. Although these findings require independent validation, they suggest that identifying insufficient evidence does not guarantee that unsupported assertions will be prevented from appearing in generated responses.

The study therefore highlights an important distinction between classification correctness, evidence verification, and the enforcement of verification judgments. A system may correctly identify a suspicious message while providing explanations that are not fully justified by the available evidence.

Overall, the findings suggest that claim-level verification offers a potentially useful mechanism for identifying weaknesses in AI-generated assessments, but its effectiveness depends on how consistently those judgments influence the final response. Future research should investigate stronger evidence-enforcement mechanisms, independently validated claim-level evaluation, and the trade-offs between reliability and computational cost.

This work contributes an empirical evaluation of claim-level verification within a retrieval-augmented phishing assessment system and identifies limitations that warrant further investigation in the development of trustworthy language model applications.

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